

Safety of Pars Plana Vitrectomy in Eyes With Plaque-Irradiated Posterior Uveal Melanoma

Alok S. Bansal, MD; Carlos G. Bianciotto, MD; Joseph I. Maguire, MD; Carl D. Regillo, MD; Jerry A. Shields, MD; Carol L. Shields, MD

Objective: To determine the long-term safety of pars plana vitrectomy (PPV) in eyes with plaque-irradiated posterior uveal melanoma.

Methods: In this retrospective case series, patients with plaque-irradiated posterior uveal melanoma subsequently underwent PPV for vitreous hemorrhage. The main outcome measures are the rates of intraocular melanoma dissemination, extrascleral extension of melanoma, local melanoma recurrence, and systemic melanoma metastasis after PPV.

Results: Forty-seven eyes of 47 patients underwent PPV for vitreous hemorrhage after iodine 125-labeled plaque radiotherapy for choroidal melanoma. The mean interval between the onset of vitreous hemorrhage and PPV was 13 (median, 10; range, 0-52) months. The mean time from PPV to last follow-up was 5 (range, 0.5-16) years.

There were no cases of intraocular melanoma dissemination or extrascleral extension of melanoma. One patient (2%) developed local choroidal melanoma recurrence (2 years after PPV and 5 years after initial plaque radiotherapy) and was successfully managed with transpupillary thermotherapy. Systemic melanoma metastasis occurred in 4 patients (9%) during a mean interval of 5 years after plaque radiotherapy. During follow-up, 43 patients (91%) were alive without systemic metastasis and 4 patients (9%) were alive with metastasis.

Conclusion: Management of vitreous hemorrhage by PPV in eyes with previously irradiated uveal melanoma appears to be safe and without increased risk for intraocular, local, orbital, or systemic dissemination of the tumor.

Arch Ophthalmol. 2012;130(10):1285-1290

Author Affiliations: Retina Service (Drs Bansal, Maguire, and Regillo) and Ocular Oncology Service (Drs Bianciotto, J. A. Shields, and C. L. Shields), Wills Eye Institute, Thomas Jefferson University, Philadelphia, Pennsylvania.

INDICATIONS FOR PARS PLANA vitrectomy (PPV) have expanded greatly since its inception in the 1970s.¹ Currently, PPV is most commonly performed for macular disease, rhegmatogenous retinal detachment, and complications of diabetic retinopathy.² Occasionally, intraocular malignant neoplasms can initiate vitreous hemorrhage (VH) before or after therapy, and PPV is considered. Concerns regarding PPV in such eyes include intraocular tumor dissemination, extraocular extension, orbital invasion, and remote spread. Some reports have indirectly commented on outcomes of PPV in eyes with treated posterior uveal melanoma.³⁻⁷ Foster et al⁸ specifically examined the safety of PPV in 9 eyes with treated posterior uveal melanoma and a mean follow-up of 2 years; one had intraocular tumor dissemination, and none experienced extrascleral extension or a metastatic event. Herein, we examine our experience regarding the long-term safety of PPV in a cohort of 47 patients undergoing PPV for VH after plaque radiotherapy for posterior uveal melanoma.

METHODS

After approval by the Wills Eye Institute institutional review board, we reviewed the medical records of patients with posterior uveal melanoma who were treated with plaque radiotherapy from June 1, 1978, through April 30, 2006, at the Ocular Oncology Service of the Wills Eye Institute, Thomas Jefferson University. Those patients who subsequently underwent PPV for VH after tumor treatment were selected for further analysis.

Baseline patient data included age, race, sex, and history of diabetes mellitus. Baseline tumor data recorded included tumor tissue (choroidal or ciliochoroidal), maximum tumor diameter (based on indirect ophthalmoscopy and B-scan ultrasonography), initial tumor thickness (based on A- and B-scan ultrasonography), retinal invasion, and extraocular extension. All tumors were treated with iodine 125-labeled plaque radiotherapy with or without adjuvant transpupillary thermotherapy; the dose of radiation to the tumor apex was recorded.

The intervals from plaque radiotherapy to the development of VH and from VH to PPV were recorded. Visual acuity and tumor thickness before PPV were recorded. The cause of the VH was determined by indirect ophthalm-

Table 1. Patient Demographics Before Melanoma Treatment in 47 Patients

Demographic Characteristic	Value ^a
Age, y	
Mean	57
Median (range)	61 (12-81)
Race	
White	45 (96)
Hispanic	2 (4)
African American	0
Asian	0
Sex	
Male	29 (62)
Female	18 (38)
History of diabetes mellitus	12 (26)

^aData are expressed as number (percentage) of patients unless otherwise indicated.

Table 2. Tumor Characteristics Before Melanoma Treatment in 47 Patients

Characteristic	Value ^a
Melanoma location	
Choroid	45 (96)
Ciliochoroidal	2 (4)
Melanoma largest base, mm	
Mean	10
Median (range)	11 (6-19)
Melanoma thickness, mm	
Mean	5
Median (range)	4.5 (2-11)
Associated features	
Retinal invasion	7 (15)
Extrascleral extension	0
Treatment	
Iodine 125-labeled plaque radiotherapy	47 (100)
Radiation dose to apex, Gy	
Mean	91
Median (range)	85 (80-156)
Adjuvant TTT	22 (47)

Abbreviation: TTT, transpupillary thermotherapy.

^aData are expressed as number (percentage) of eyes unless otherwise indicated.

moscopy, fundus photography, and B-scan ultrasonography. All eyes underwent standard 3-port PPV.

The primary outcomes of this analysis were the rates of intraocular melanoma dissemination, extrascleral extension of melanoma, local melanoma recurrence, and systemic melanoma metastasis after PPV. Secondary outcomes included resolution of VH, visual acuity reduction or improvement, development of rhegmatogenous retinal detachment or endophthalmitis, and the need for enucleation after PPV.

RESULTS

From June 1, 1978, through April 30, 2006, 3841 patients (3841 eyes) underwent plaque radiotherapy for uveal melanoma at the Oncology Service of the Wills Eye Institute; from this cohort, 47 eyes of 47 patients subsequently underwent PPV for VH after tumor treatment from May 1, 1992, through August 31, 2005. Patient demo-

Table 3. Safety of PPV in 47 Eyes With Irradiated Posterior Uveal Melanoma: Features at Time of Vitrectomy

Feature	Value
Interval from VH to PPV, mo	
Mean	13
Median (range)	10 (0-52)
Visual acuity before PPV, No. (%) ^a	
20/20-20/40	0
20/60-20/100	3 (6)
≤20/200	43 (91)
Melanoma thickness, mm	
Mean	3
Median (range)	3 (0-6)

Abbreviations: PPV, pars plana vitrectomy; VH, vitreous hemorrhage.

^aInformation on visual acuity was absent in 1 case.

Table 4. Safety of PPV in 47 Eyes With Irradiated Posterior Uveal Melanoma: Outcomes After Vitrectomy

Outcome	Value ^a
Follow-up after PPV, y	
Mean	5
Median (range)	3 (0.5-16)
Primary outcomes	
Intraocular melanoma dissemination	0
Extrascleral extension of melanoma	0
Local melanoma recurrence	1 (2)
Systemic melanoma metastasis	4 (9)
Resolution of VH (secondary outcome)	35 (74)
Visual acuity (n = 41)	
20/20-20/40	3 (6)
20/60-20/100	4 (9)
≤20/200	34 (72)
Gained ≥1 lines	19 (40)
Rhegmatogenous retinal detachment	2 (4)
Endophthalmitis	0
Enucleation	6 (13)
Neovascular glaucoma	7 (15)
Patient status at last visit	
Alive without metastasis	43 (91)
Alive with metastasis	4 (9)
Deaths due to other causes	0
Deaths due to melanoma	0

Abbreviations: PPV, pars plana vitrectomy; VH, vitreous hemorrhage.

^aData are expressed as number (percentage) of eyes unless otherwise indicated.

graphics at the initial examination before tumor treatment are listed in **Table 1**. The baseline tumor characteristics and treatment are listed in **Table 2**. All eyes had choroidal or ciliochoroidal tumors. The mean tumor base was 10 mm, and the mean tumor thickness was 5 mm before tumor treatment. Retinal invasion was present in 7 eyes (15%). No patients underwent fine-needle aspiration biopsy before plaque treatment. All eyes were treated with ¹²⁵I-labeled plaque radiotherapy with median tumor apex dose of 85 Gy without any intraoperative complications.

The mean interval between plaque radiotherapy and the development of VH was 22 (median, 15; range, 0-137) months. The most prevalent causes of VH included presumed tumor necrosis in 21 eyes (45%), proliferative ra-

Table 5. Safety of PPV in Eyes With Irradiated Posterior Uveal Melanoma: Summary of Literature

Source	No. of Eyes	Tumor Treatment	Indication for PPV	Mean Follow-up After PPV, y ^a	No. of Eyes			
					Intraocular Tumor Dissemination	Extrasceral Extension	Local Recurrence	Metastasis
Radtke et al, ³ 1991	1	Plaque ^b	ERD	3	0	0	0	0
Haimovici et al, ⁴ 1996	1	Plaque ^b	RRD	3	0	0	0	0
Mashayekhi et al, ⁵ 2008	7	Plaque ^b /TTT	RRD	4 (0.5-8)	0	0	1	1
Ferreira et al, ⁶ 2008	1	Plaque ^b	VH	1	0	0	0	0
Song et al, ⁷ 2010	3	Gamma knife	ERD	NA	0	0	0	0
Foster et al, ⁸ 2003	9	Plaque ^b /TTT	VH, ERM, RRD, MH	2 (0.5-2)	1	0	0	0
Present study	47	Plaque ^b /TTT	VH	5 (0.5-16)	0	0	1	4

Abbreviations: ERD, exudative retinal detachment; ERM, epiretinal membrane; MH, macular hole; NA, not available; PPV, pars plana vitrectomy; RRD, rhegmatogenous retinal detachment; TTT, transpupillary thermotherapy; VH, vitreous hemorrhage.

^aRange is included where available.

^bIndicates iodine 125- or ruthenium 106-labeled plaque radiotherapy.

diation retinopathy in 13 (28%), posterior vitreous detachment in 7 (15%), and retinal break in 2 (4%). No eyes had a rhegmatogenous retinal detachment. Initial management included observation in 34 eyes (72%) and pan-retinal photocoagulation in 11 (23%). Recurrent VH occurred in 15 eyes (32%) before PPV.

The mean interval between onset of VH and PPV was 13 (median, 10; range, 0-52) months, and the mean interval between the plaque radiotherapy and PPV was 35 (median, 30; range, 6-137) months. The ocular features at the time of PPV are listed in **Table 3**. Visual acuity before PPV was 20/60 to 20/100 in 3 patients (6%) and 20/200 or worse in 43 (91%) (data were missing in 1 patient). In 45 eyes (96%), convincing ultrasonographic reduction in melanoma thickness was detected after plaque radiotherapy, with minimal reduction in 2 (4%). Results of vitreous cytology were available in 4 patients; findings were negative for malignant cells in 3 and equivocal in 1.

The primary outcomes analysis is listed in **Table 4**. The mean time from PPV to the last follow-up was 5 years. No cases of intraocular melanoma dissemination or extrasceral extension of melanoma were found after PPV. Only 1 patient developed local melanoma recurrence within the choroid (2 years after PPV and 5 years after plaque radiotherapy), which was successfully managed with transpupillary thermotherapy. This patient had had 2 recurrences before PPV, both of which were managed with transpupillary thermotherapy also. Four patients (9%) developed systemic melanoma metastasis after PPV. The mean time from plaque radiotherapy to metastasis was 5 years; from PPV to metastasis, 3 years. Metastasis occurred in the liver (n = 2), lung (n = 1), and multiple sites (n = 1).

The secondary outcomes are listed in Table 4. Permanent resolution of the VH occurred in 35 eyes (74%). Visual acuity was 20/20 to 20/40 in 3 patients (6%), 20/60 to 20/100 in 4 (9%), and 20/200 or worse in 34 (72%). Poor visual acuity was generally due to adverse effects of radiation therapy. Enucleation was necessary in 6 cases (13%) for a blind, painful eye; mean time from PPV to enucleation was 8 months. We found no histopathologic evidence of extrasceral tumor dissemination. After PPV, 7 patients (15%) developed neovascular glaucoma that was managed with medications (n = 3), an aqueous shunt device (n = 1), re-

peated PPV (n = 1), or enucleation (n = 2). At last follow-up, 43 patients (91%) were alive without systemic metastasis and 4 patients (9%) were alive with metastasis. There were no deaths due to melanoma or other causes.

COMMENT

Adverse effects of plaque radiotherapy for posterior uveal melanoma include maculopathy,⁹ papillopathy,¹⁰ radiation retinopathy,¹¹ proliferative retinopathy,¹² and VH,¹³ among others.^{14,15} Gündüz et al⁹ reviewed complications of radiation therapy in 630 eyes with plaque-irradiated melanoma and described the 10-year risk for maculopathy (59%), papillopathy (23%), and VH (19%). Multivariate analysis revealed factors predictive of VH in uveal melanoma included mushroom shape and being older than 60 years. Bianciotto et al¹³ specifically evaluated 403 eyes with VH after plaque radiotherapy for uveal melanoma and found the causes of VH included tumor necrosis (29%), proliferative radiation retinopathy (24%), posterior vitreous detachment (16%), retinal venous occlusion (5%), and retinal invasion (3%). Multivariate analysis in that study demonstrated baseline diabetic retinopathy, shorter distance to the optic nerve, thicker tumors, and Bruch membrane rupture as additional risk factors for the development of VH. In addition, the report advised that VH be observed for spontaneous clearance for 3 to 6 months because that occurred in 48% of eyes. However, in eyes with severe, persistent, or recurrent VH and those associated with neovascularization of the iris or retina, PPV is considered. Benefits of PPV include not only improved visual acuity and prevention of complications of neovascularization, but of equal importance, enhanced local intraocular tumor surveillance.

Little data have been published regarding the safety of vitreous surgical procedures in eyes with posterior uveal melanoma. There have been few reports¹⁶⁻¹⁹ on PPV in eyes with unsuspected intraocular neoplasms later found to harbor uveal melanoma. Of 6 such eyes reported in the literature, 4 eventually had intraocular tumor dissemination based on enucleated specimens and 2 demonstrated evidence of extrasceral extension; 1 patient died

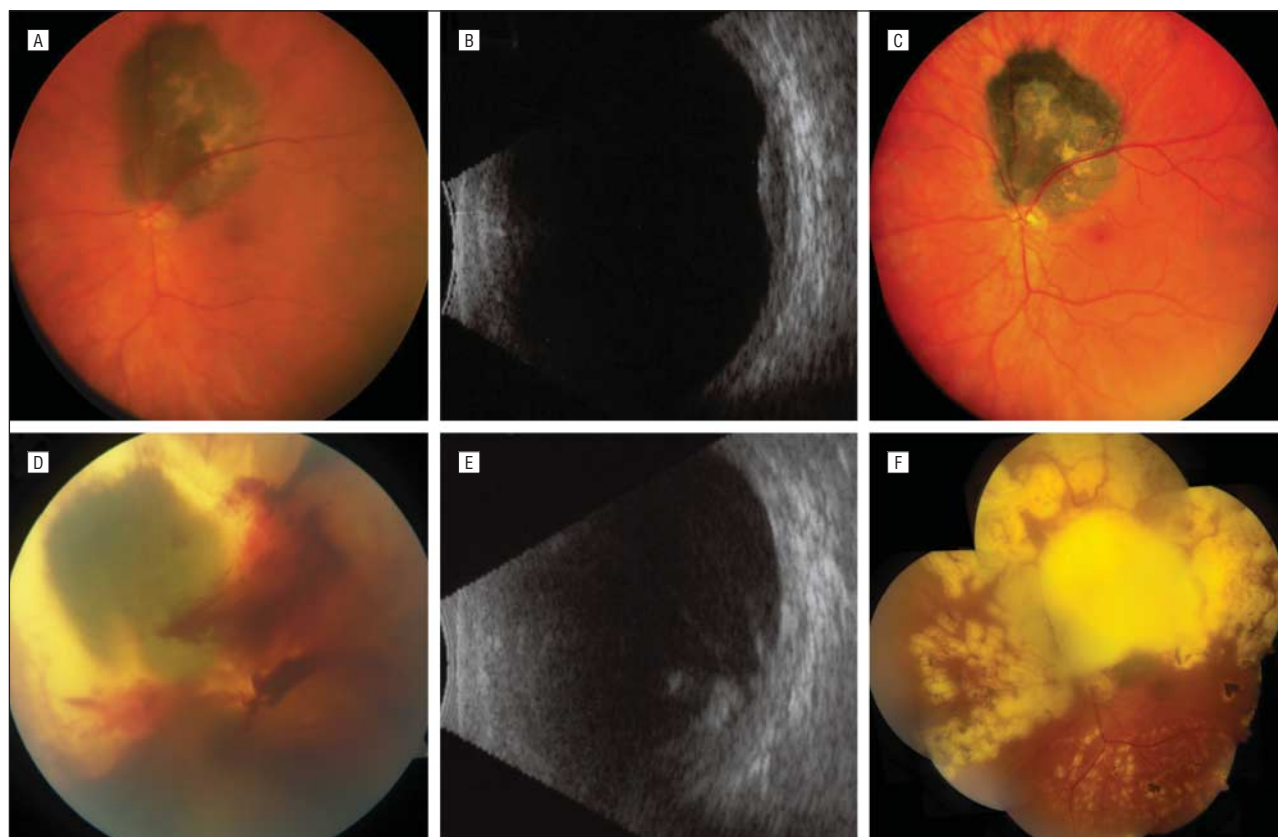


Figure 1. A 43-year-old woman with juxtapapillary choroidal melanoma. A, Color fundus photograph before treatment. B, Ultrasonogram measuring tumor thickness of 2.3 mm. C, Color fundus photograph 5 months after iodine 125-labeled plaque radiotherapy showing slight regression of the tumor and stable margins. D, Color fundus photograph showing development of recurrent vitreous hemorrhage (VH) 17 months after tumor treatment. E, Ultrasonogram obtained after observation of the VH for 12 months, at which time the tumor thickness measured 1.5 mm before pars plana vitrectomy (PPV). F, Color fundus photograph 5 years after PPV and sector photocoagulation showing an atrophic scar; the patient was alive and well without evidence of metastasis.

of liver metastasis.¹⁶⁻¹⁹ In general, melanoma should be ruled out via ultrasonography and/or other testing in any case of a VH of unknown origin.

In eyes that have a clinically suspected uveal melanoma, biopsy of tumor tissue with fine-needle aspiration is increasingly used for prognostic purposes. Shields et al²⁰ reported a series of 500 eyes with uveal melanoma undergoing fine-needle aspiration–sampled genetic testing and found no evidence of intraocular or extraocular tumor dissemination. A review of nearly 2000 fine-needle aspiration biopsies during the past 35 years found no cases of intraocular melanoma dissemination or local melanoma recurrence. Sampling by PPV is more invasive and is used less frequently. Five studies with a total of 35 patients have reported safety outcomes after PPV with transretinal biopsy in eyes with untreated melanoma.²¹⁻²⁵ After histopathologic confirmation of melanoma, these eyes underwent treatment with plaque radiotherapy (13 eyes), enucleation (11 eyes), proton-beam irradiation (10 eyes), or transscleral resection (1 eye). No cases of intraocular melanoma dissemination, extrascleral extension of melanoma, or local melanoma recurrence were detected. However, Bechrakis et al²² reported that 3 of 13 patients with biopsy-confirmed melanoma developed systemic liver metastasis.

Concerning the safety of PPV in eyes with treated posterior uveal melanoma, data in the literature are also limited (**Table 5**). Foster et al⁸ reported 9 eyes that had PPV

after treatment of uveal melanoma, and 1 patient developed intraocular tumor dissemination that resulted in enucleation. None of the eyes developed extrascleral extension. At mean 2-year follow-up, all 9 patients were alive without metastasis. Radtke et al³ reported a case of exudative retinal detachment after plaque treatment for choroidal melanoma managed successfully with PPV without tumor dissemination at 37 months after PPV. Haimovici et al⁴ described a patient who underwent successful PPV for rhegmatogenous retinal detachment after plaque treatment for choroidal melanoma without tumor dissemination at 39 months after PPV. Mashayekhi et al⁵ reported 7 eyes that underwent PPV for rhegmatogenous retinal detachment after plaque or transpupillary thermotherapy for posterior uveal melanoma. In that series, 1 patient developed a local tumor recurrence, and 1 patient developed systemic metastasis. Ferreyra et al⁶ described a patient who underwent PPV with endoresection of a previously irradiated choroidal melanoma for treatment of VH and glaucoma. This patient had undergone 5 prior vitrectomies and did not develop extrascleral extension or systemic metastasis 12 months after PPV. Song et al⁷ recently described 3 patients who underwent PPV for persistent exudative retinal detachment after gamma knife treatment for choroidal melanoma. No cases of intraocular melanoma dissemination, extrascleral extension of melanoma, local melanoma recurrence, or systemic melanoma metastasis were found.

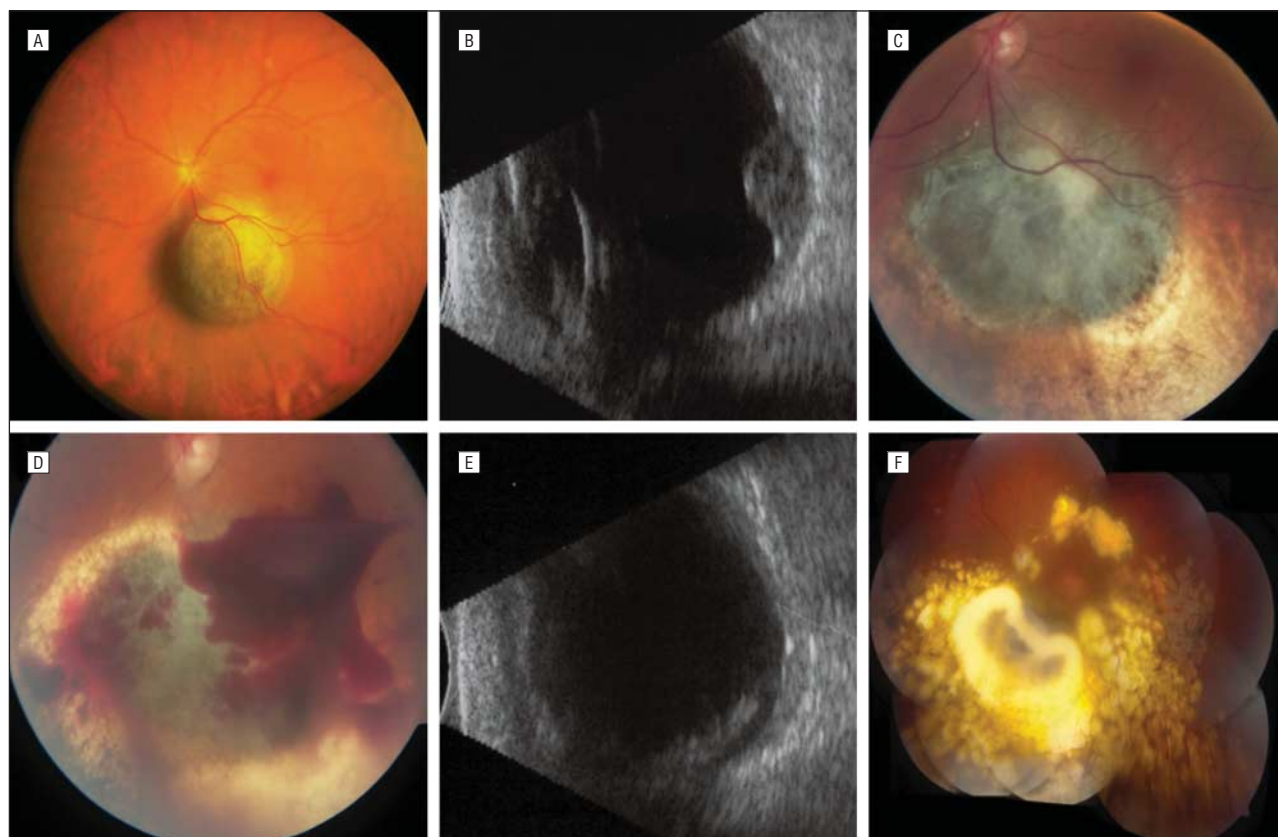


Figure 2. A 46-year-old man with choroidal melanoma. A, Color fundus photograph before treatment. B, Ultrasonogram measuring tumor thickness of 5.0 mm. C, Color fundus photograph 10 months after iodine 125-labeled plaque radiotherapy showing regression of the tumor. D, Color fundus photograph showing development of persistent vitreous hemorrhage (VH) 15 months after tumor treatment. E, Ultrasonogram obtained after observation of the VH for 3 months, at which time tumor thickness measured 1.8 mm before pars plana vitrectomy (PPV). F, Color fundus photograph 5 years after PPV and sector photocoagulation showing an atrophic scar; the patient was alive and well without evidence of metastasis.

Based on our data and a review of the literature, PPV for VH after plaque radiotherapy of uveal melanoma appears to be safe and without increased risk for local or systemic spread. When comparing our patients with a published database of more than 8000 affected patients with uveal melanoma, systemic metastasis was not increased. In our cohort, 9% had metastasis vs 15% in the large published cohort at mean follow-up of 5 years.²⁶

Secondary outcomes of our study found PPV to be successful in resolving VH in 74% of cases, and visual acuity improved by 1 line or more in 19 eyes (40%). Reasons for poor visual outcome included radiation maculopathy in 38 eyes (81%) and radiation papillopathy in 26 (55%). Although glaucoma drainage implants are generally not indicated in eyes harboring untreated melanoma, 1 patient in this series had an aqueous shunt device implanted at another institution for neovascular glaucoma more than 2 years after plaque therapy and PPV. Given the long interval between the tumor treatment and drainage implant and the lack of viable melanoma cells from the vitreous sampling, we believe this procedure to be safe. In addition, in this series, 6 eyes underwent enucleation after PPV. No histopathologic evidence of tumor dissemination within the globe or orbit was found in these patients.

Several limitations are inherent in our retrospective case series. First, selection bias is a potential shortcoming because we only included patients who underwent PPV for VH in an eye that showed regressed melanoma.

Other indications for vitreous surgery in eyes with treated uveal melanoma were not evaluated herein. Second, several vitreoretinal surgeons in our institution and elsewhere performed PPV using a standard technique but with different preferences. The lack of negative results, despite the multiplicity of vitreous surgeons, argues for the safety of this procedure in eyes undergoing ¹²⁵I-labeled plaque irradiation. Last, vitreous cytology findings were available in only 4 eyes. Our preference would have been to sample the vitreous in all eyes, but pathological evaluation of vitreous samples is not performed at all institutions. Nevertheless, all patients in this series had documented tumor stability before vitrectomy.

In summary, based on our results and a review of the literature, PPV in eyes with treated posterior uveal melanoma appears to be safe (**Figure 1** and **Figure 2**) and does not increase the risk of intraocular melanoma dissemination, extrascleral extension of melanoma, local melanoma recurrence, or systemic melanoma metastasis. Larger cohorts with longer follow-up would provide more robust data.

Submitted for Publication: April 21, 2012; final revision received May 27, 2012; accepted May 29, 2012.

Correspondence: Carol L. Shields, MD, Ocular Oncology Service, Wills Eye Institute, Thomas Jefferson University, 840 Walnut St, Ste 1440, Philadelphia, PA 19107.

Author Contributions: Dr C. L. Shields had full access to

all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Financial Disclosure: None reported.

Funding/Support: This study was supported by the J. Arch McNamara Research Fund (Dr Bansal) and the Eye Tumor Research Foundation (Drs J. A. Shields and C. L. Shields).

Previous Presentation: This material was previously presented as a poster at the American Academy of Ophthalmology; October 23, 2011; Orlando, FL.

REFERENCES

1. Machemer R, Buettner H, Norton EW, Parel JM. Vitrectomy: a pars plana approach. *Trans Am Acad Ophthalmol Otolaryngol*. 1971;75(4):813-820.
2. Gupta OP, Ho AC, Kaiser PK, et al. Short-term outcomes of 23-gauge pars plana vitrectomy. *Am J Ophthalmol*. 2008;146(2):193-197.
3. Radtke ND, Augsburger JJ, Schmitt T. Management of exudative retinal detachment after plaque therapy for intraocular melanoma. *Am J Ophthalmol*. 1991;112(1):92-94.
4. Haimovici R, Mukai S, Schachat AP, et al. Rhegmatogenous retinal detachment in eyes with uveal melanoma. *Retina*. 1996;16(6):488-496.
5. Mashayekhi A, Shields CL, Lee SC, Marr BP, Shields JA. Retinal break and rhegmatogenous retinal detachment after transpupillary thermotherapy as primary or adjunct treatment of choroidal melanoma. *Retina*. 2008;28(2):274-281.
6. Ferreyra HA, Goldbaum MH, Weinreb RN. Endoresection of irradiated choroidal melanoma as a treatment for intractable vitreous hemorrhage and secondary blood-induced glaucoma. *Semin Ophthalmol*. 2008;23(2):135-138.
7. Song WK, Yang WI, Byeon SH, Koh HJ, Kwon OW, Lee SC. Clinicopathologic report of uveal melanoma with persistent exudative retinal detachment after gamma knife radiosurgery. *Ophthalmologica*. 2010;224(1):16-21.
8. Foster WJ, Harbour JW, Holekamp NM, Shah GK, Thomas MA. Pars plana vitrectomy in eyes containing a treated posterior uveal melanoma. *Am J Ophthalmol*. 2003;136(3):471-476.
9. Gündüz K, Shields CL, Shields JA, Cater J, Freire JE, Brady LW. Radiation complications and tumor control after plaque radiotherapy of choroidal melanoma with macular involvement. *Am J Ophthalmol*. 1999;127(5):579-589.
10. Shields CL, Demirci H, Marr BP, et al. Intravitreal triamcinolone acetonide for acute radiation papillopathy. *Retina*. 2006;26(5):537-544.
11. Gündüz K, Shields CL, Shields JA, Cater J, Freire JE, Brady LW. Radiation retinopathy following plaque radiotherapy for posterior uveal melanoma. *Arch Ophthalmol*. 1999;117(5):609-614.
12. Bianciotto C, Shields CL, Pirondini C, Mashayekhi A, Furuta M, Shields JA. Proliferative radiation retinopathy after plaque radiotherapy for uveal melanoma. *Ophthalmology*. 2010;117(5):1005-1012.
13. Bianciotto C, Shields CL, Pirondini C, Mashayekhi A, Furuta M, Shields JA. Vitreous hemorrhage after plaque radiotherapy for uveal melanoma. *Retina*. 2012;32(6):1156-1164.
14. Shields CL, Cater J, Shields JA, et al. Combined plaque radiotherapy and transpupillary thermotherapy for choroidal melanoma: tumor control and treatment complications in 270 consecutive patients. *Arch Ophthalmol*. 2002;120(7):933-940.
15. Shields CL, Shields JA, Cater J, et al. Plaque radiotherapy for uveal melanoma: long-term visual outcome in 1106 consecutive patients. *Arch Ophthalmol*. 2000;118(9):1219-1228.
16. Laqua H, Völcker HE. Pars plana vitrectomy in eyes with malignant melanoma. *Graefes Arch Clin Exp Ophthalmol*. 1983;220(6):279-284.
17. Traboulsi EI, Jalkh AE, Frangieh GT, Tomb J. Vitreous histocytology of primary choroidal malignant melanoma. *Ann Ophthalmol*. 1987;19(2):45-47.
18. DiSclafani M, Lederman IR. Intraocular malignant melanoma simulating uveal prolapse. *Ann Ophthalmol*. 1988;20(2):68-70.
19. Kavanagh MC, Everman KR, Opremac EM, Foster JA. Uveal melanoma with massive extrascleral extension via pars plana vitrectomy sites. *Ophthalm Plast Reconstr Surg*. 2008;24(4):334-336.
20. Shields CL, Ganguly A, Bianciotto CG, Turaka K, Tavallali A, Shields JA. Prognosis of uveal melanoma in 500 cases using genetic testing of fine-needle aspiration biopsy specimens. *Ophthalmology*. 2011;118(2):396-401.
21. Fastenberg DM, Finger PT, Chess Q, Koizumi JH, Packer S. Vitrectomy retinotomy aspiration biopsy of choroidal tumors. *Am J Ophthalmol*. 1990;110(4):361-365.
22. Bechrakis NE, Foerster MH, Bornfeld N. Biopsy in indeterminate intraocular tumors. *Ophthalmology*. 2002;109(2):235-242.
23. Kvant A, Seregard S, Kopp ED, All-Ericsson C, Landau I, Berglin L. Choroidal biopsies for intraocular tumors of indeterminate origin. *Am J Ophthalmol*. 2005;140(6):1002-1006.
24. Gibran SK, Kapoor KG. Management of exudative retinal detachment in choroidal melanoma. *Clin Experiment Ophthalmol*. 2009;37(7):654-659.
25. Abi-Ayad N, Grange JD, Salle M, Kodjikian L. Transretinal uveal melanoma biopsy with 25-gauge vitrectomy system [published online May 19, 2011]. *Acta Ophthalmol*. <http://onlinelibrary.wiley.com/doi/10.1111/j.1755-3768.2011.02172.x/abstract>. Accessed January 16, 2012.
26. Shields CL, Furuta M, Thangappan A, et al. Metastasis of uveal melanoma millimeter-by-millimeter in 8033 consecutive eyes. *Arch Ophthalmol*. 2009;127(8):989-998.